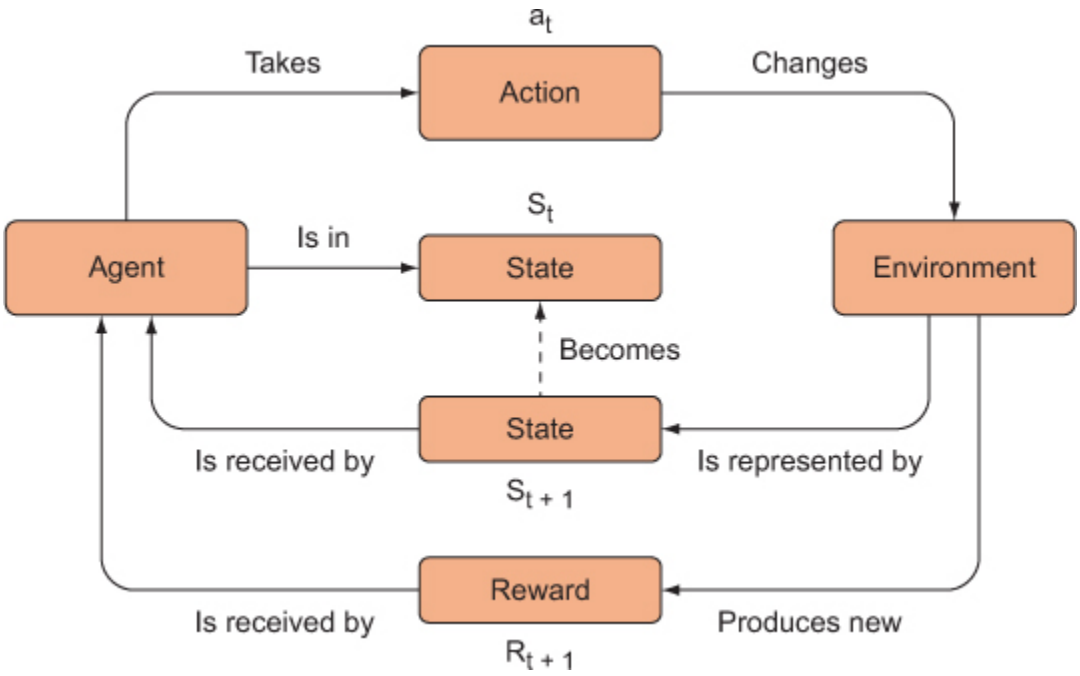
**Figure 2.1. The standard reinforcement learning model in which an agent takes actions in an evolving environment that produces rewards to reinforce the actions of the agent.**




More generally, string diagrams (sometimes referred to as *wiring diagrams* in other sources) are flow-like diagrams that represent the flow of typed data along strings (i.e., directed or undirected arrows) into processes (computations, functions, transformations, processes, etc.), which are represented as boxes. The important difference between string diagrams and other similar-looking flow diagrams you may have seen is that all the data on the strings is explicitly typed (e.g., a numpy array with shape `[10, 10]`, or maybe a floating-point number), and the diagrams are fully compositional. By compositional, we mean that we can zoom in or out on the diagram to see the bigger more abstract picture or to drill down to the computational details.

If we're showing a higher-level depiction, the process boxes may just be labeled with a word or short phrase indicating the kind of process that happens, but we could also show a zoomed-in view of that process box that reveals all its internal details, composed of its own set of substrings and subprocesses. The compositional nature of these diagrams also means that we can plug parts of one diagram into another diagram, forming more